Do Now · retrieval

C# Login Project · Start from Scratch

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

A rising-edge-triggered D-type flip-flop has Q = 0 and D = 1. What happens to Q when the next rising edge occurs?

Q2

Why do we group neighbouring 1s together on a Karnaugh map?

Q3

Push A, then B, then C onto an empty stack. Pop once and then push D. Which item is at the top, and what is the remaining stack from top to bottom?

Q4

Using 8-bit two’s complement, write −13 and explain why simply putting 1 before the 7-bit magnitude is not correct.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

What must a login program do? — Read two inputs, decide whether both match the demonstration account, and show a clear result. Design the sequence before typing code.

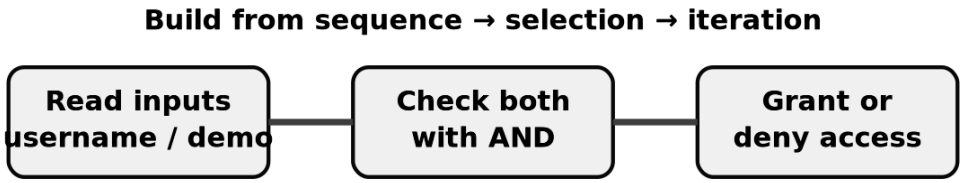
Core idea 2

Why start with sequence? — Read username; read password; display what was entered only during debugging. In the final version never echo a password.

Core idea 3

Introduce selection — Use AND (&&) to require BOTH username and password to match. One wrong value must mean access denied.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Make a second attempt possible — A loop can repeat after invalid credentials. Add a maximum number of attempts to avoid an endless loop.

Core idea 5

Data-structure revision — Write one accurate four-option MCQ each on arrays (up to 3D), records, lists and tuples. Add answers and distractor explanations.

Vocabulary · use the correct technical terms

Statements executed in order. — Statements executed in order.

Branching with if/else. — Branching with if/else.

Repeating statements with a loop. — Repeating statements with a loop.

Checking input meets defined rules. — Checking input meets defined rules.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Model the first three lines — not the finished program

Follow the worked method

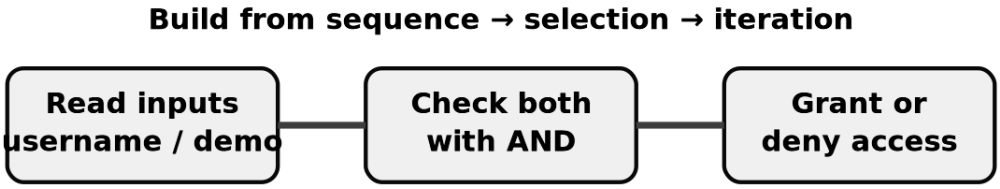
1. Create a console program — CODE:
using System;

Console.Write("Username: ");
string username = Console.ReadLine() ?? "";

2. Add the second input — CODE:
Console.Write("Demo password: ");
string password = Console.ReadLine() ?? "";

3. Plan, then add selection — CODE:
// Write an if statement:
// username must equal "student"
// AND password must equal "demo123"
// Print "Access granted" or "Access denied".

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain your solution first
What Boolean operator makes BOTH checks compulsory?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

CODING 1 · SEQUENCE

From a blank C# console program, request a username and a demonstration password. Use two string variables. Do not copy in a finished solution.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

SYNOPTIC MCQ BANK

Write FOUR original four-option MCQs: one on arrays up to 3D, one on records, one on lists and one on tuples. Mark the correct letter, explain it and justify each distractor.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A login should allow access when the username is correct even if the password is wrong.

Use key vocabulary from memory

1. Explain Statements executed in order. in your own words.

2. Explain Branching with if/else. in your own words.

3. Explain Repeating statements with a loop. in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.